1. Open the standard model “gttft1” in the Manage Models app.

2. Click **Edit** on the top right.

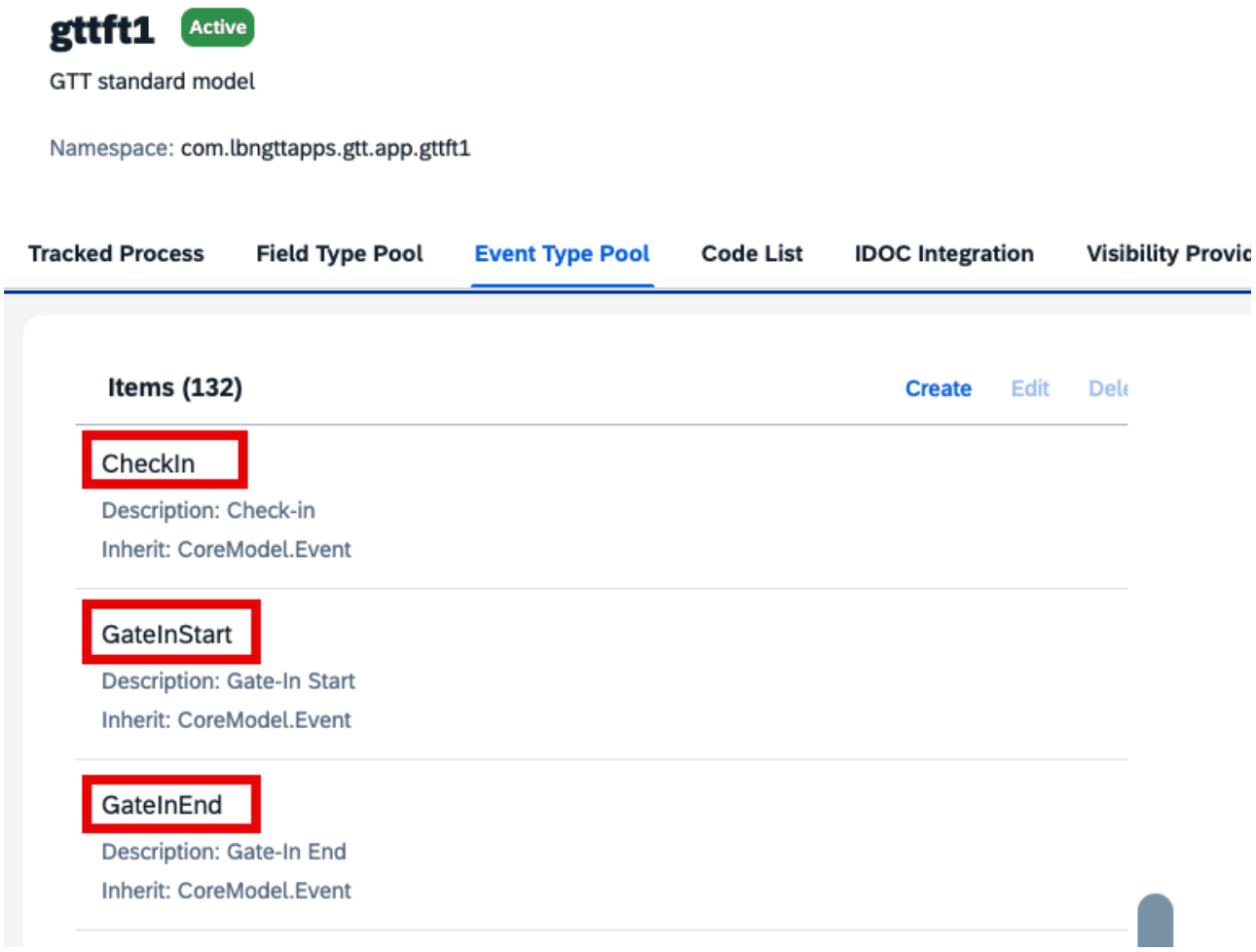
3. Switch to the **Event to Action** tab and in the **User Script Before Standard Script**, add the variable "GTT_TS_EVENT_TYPES_IN_EXEC" with the list of event types in the following format"

   ### Example

   GTT_TS_EVENT_TYPES_IN_EXEC= [“Departure”, “Arrival”, “OnTimeEvent”, "ZZTest"].

   Note the following:

   - Custom event types are also supported but they must be added to the **Event Type Pool** and the **Admissible Planned/Unplanned Event List** of the “Shipment”, “Freight Unit” and "Tracking Unit" tracked processes. For more information, see the following:
     - Create Event Type
     - Create Admissible Planned Event
     - Create Admissible Unplanned Event
   - You must enter the name of the event type instead of its description, as indicated in the following screenshot. It's case sensitive. You don't need to add the model namespace before the event type:



   - If you want to add more event types, remember to copy the default list and paste them here first, as the custom list you are creating now will override the default list instead of complementing it.

4. Save the change and deploy the model.

Supported Event Types

By default, the following list of event types are specified for the variable in the **Standard Declaration**:

- Departure
- Arrival
- LoadingStart
- LoadingEnd
- Unloading Start
- Unloading End
- POPU
- POD
- CheckIn
- Coupling
- Decoupling
- GateInStart
- GateInEnd
- GateOutStart
- GateOutEnd
- GateInEmpty
- GateInFull
- GateOutEmpty
- GateOutFull
- Stuffing
- Unstuffing
- OutForDelivery
- CustomHoldStart
- CustomHoldEnd
- ReadyForPickup
- PickupEnd
- Handover
- CrossDockHandover
- ReadyForCarriage
- Receive
- ReceivedFromCarrier

- GoodsReceipt
- Delivered
- Return
- ManifestReady
- ReceivedFromShipper
- ConsigneeNotified

Stage-Level Observation Between Tracked Processes

Introduction

When a tracked process observes another tracked process at stage level, the event correlation follows the following rules:

- For actual events that can match a planed event in the observing tracked process, they can be correlated to the observing tracked process from the observed tracked process.
- For actual events that cannot match a planned event in the observing tracked process, only those of the observed stages can be correlated to the observing tracked process from the observed tracked process. In such case, the correlation starts after the first planned event is reported for the observed tracked process, and the actual event must:
    - be correlated to the observing tracked process.
    - match a planned event of the observing tracked process.

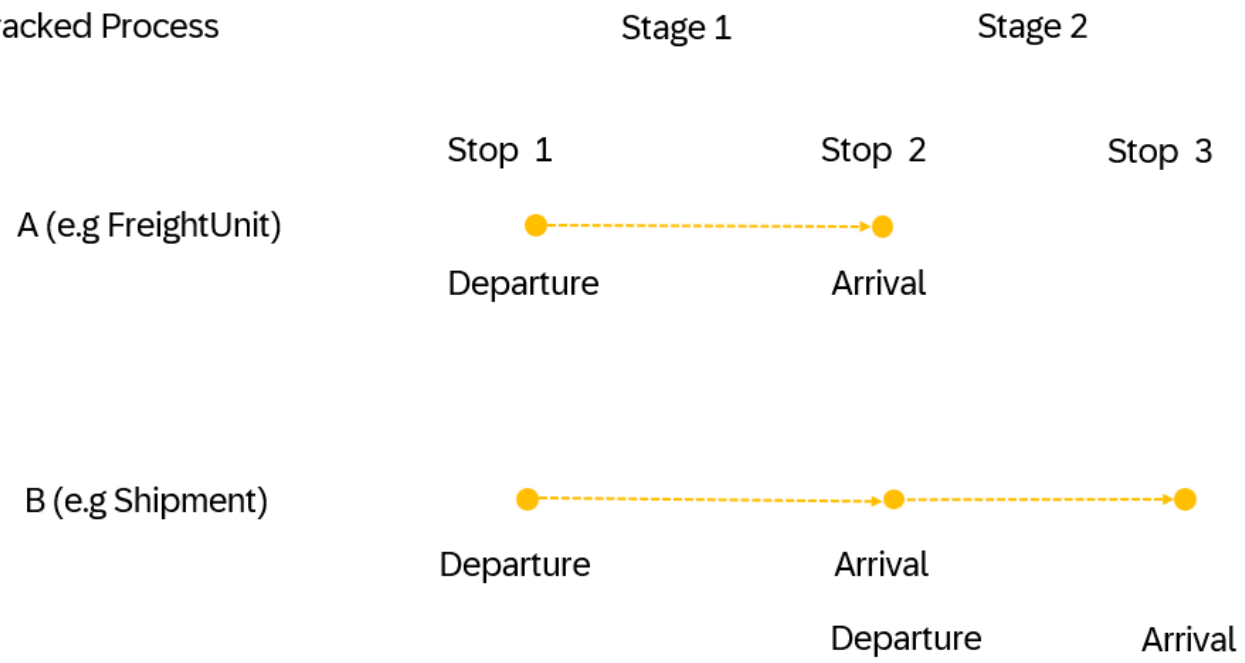
  The correlation stops once the Arrival event on the last stop of the observed stages is reported for the observed tracked process.

In the fulfillment tracking apps, those correlated events from the observed tracked process can be displayed in the timeline or events table of the observing tracked process.

The stage-level observation applies to the following observation scenarios:

- FreightUnit observes Shipment
- FreightUnit observes TrackingUnit
- Shipment observes TrackingUnit
- OutboundDelivery observes Shipment
- InboundDelivery observes Shipment

Here is an example that helps you better understand the event correlation for the stage-level observation:



In this example, tracked process A is assumed to observe tracked process B.

For tracked process A, there are two stops, each has a planned event.

For tracked process B, there are three stops. There are two planned events (planned Arrival and Departure events) on stop 2.

The observed stage is from stop 1 to stop 2.

Case 1

The Departure event is reported on stop 1 of tracked process B.

This Departure event can be correlated to tracked process A because the Departure event of tracked process B can match a planned event in tracked process A.

Case 2

After departure, an unplanned Location Update event is reported on stop 1 of tracked process B.

This unplanned Location update can be correlated from tracked process B to tracked process A, because:

- This Departure event matches the Departure event on stop 1 of tracked process A.
- The Departure event has already reported and correlated to tracked process A.

Case 3

The Arrival event on stop 2 of tracked process B is reported.

This Arrival event can be correlated from tracked process B to tracked process A. Meanwhile, tracked process A stops getting actual events that cannot match its planned events from tracked process B because its Arrival event on the last stop of the observed stage has been reported.

Case 4

The Departure event on stop 2 of tracked process B is reported.

This Departure event cannot be correlated from tracked process B to tracked process A because this Departure event cannot match a planned event of tracked process A, and this actual event is not on the observed stage.

As a result,

- All actual events on the observed stage (from stop 1- stop 2) can be correlated from tracked process B to tracked process A.
- All actual events from stop 2 to stop 3 cannot be correlated from tracked process B to tracked process A.

Deactivating Stage-Level Observation

By default, the stage-level observation is activated. You can use the flag "`gttft1_StageLevelObserveSwitchOn`" in the event-to-action script to deactivate it. As a result, the observing tracked process receives all actual events correlated from the observed tracked process.

To deactivate the stage-level observation, do the following:

1. Go to the standard model "gttft1" of the Manage Models app.
2. In the **User Script before Standard Script** section of the **Event to Action** tab, set the value of "`gttft1_StageLevelObserveSwitchOn`" to "false". By default, the value is set to "true".

Note

It only applies to the actual events reported after the model is deployed.

Once the stage-level observation is deactivated for an existing tracked process, when you set the value of the flag back to "true", it cannot be activated again.

The sample scripts for deactivating the stage-level observation are provided for your reference:

- FreightUnit observes TrackingUnit or Shipment

  Following is the sample script for deactivating the stage-level observation between freight unit and tracking unit, or between freight unit and shipment:

  ### Sample Code

  ```
  if(scriptForFreightUnit) {
  gttft1_stageLevelObserveSwitchOn = false;
  }
  ```

- Shipment observes TrackingUnit

  Following is the sample script for deactivating the stage-level observation between shipment and tracking unit:

  ### Sample Code

  ```
  if(scriptForShipment) {
  gttft1_stageLevelObserveSwitchOn = false;
  ```

```
}
```

- OutboundDelivery observes Shipment

  Following is the sample script for deactivating the stage-level observation between outbound delivery and shipment:

  Sample Code

  ```
  if(scriptForOutboundDelivery) {
  gttft1_stageLevelObserveSwitchOn = false;
  }
  ```

- InboundDelivery observes Shipment

  Following is the sample script for deactivating the stage-level observation between inbound delivery and shipment:

  Sample Code

  ```
  if(scriptForInboundDelivery) {
  gttft1_stageLevelObserveSwitchOn = false;
  }
  ```

Important Terms

You can find explanations for some concepts related to tracked processes and events.

Execution Status

Shows the execution progress of a tracked process. The following tables describe the execution status logic of different tracked processes.

Shipment

Execution Status	Description
Not Started	The initial execution status of a shipment.
In Execution	If a planned event of the shipment is reported, the execution status changes to In Execution.
Completed	When one of the conditions specific to the following event types is met, the shipment is considered completed: • **Proof of Delivery** ◦ All planned "Proof of Delivery" events on the destination are reported, or all planned "Proof of Delivery" events are reported. ◦ If there is no planned "Proof of Delivery" event on the destination, an unplanned "Proof of Delivery" event is reported on the destination. • **Delivered** ◦ All planned "Delivered" events on the destination are reported.

Execution Status	Description
	◦ If there are no planned "Delivered" events on the destination, an unplanned "Delivered" event is reported on the destination. • **Unloading End** ◦ All planned "Unloading End" events on the destination are reported. ◦ If there are no planned "Unloading End" events on the destination, an unplanned "Unloading End" event is reported on the destination. • **Goods Receipt** ◦ An unplanned "Goods Receipt" is reported. • **Arrival** ◦ If there are no planned "Proof of Delivery", "Delivered" or “Unloading End”events on the destination, all planned "Arrival" events on the destination are reported. ◦ When this shipment is assigned with tracking units: if there are no planned "Proof of Delivery" events nor "Delivered" events on the destination, all planned "Arrival" events are reported. The following case can also complete a shipment: • Its **last milestone event** is reported. Whether an actual event is the last milestone event or not is indicated by the field “**isFinal**”. If it has the field "**isFinal**" whose value is “true”, then it is the last milestone.

Freight Unit

Execution Status	Description
Not Started	The initial execution status of a freight unit.
In Execution	If a planned event of the freight unit is reported, the execution status changes to In Execution.
Completed	When one of the conditions specific to the following event types is met, the freight unit is considered completed: • **Proof of Delivery** ◦ All planned "Proof of Delivery" events on the destination are reported, or all planned "Proof of Delivery" events are reported. ◦ If there is no planned "Proof of Delivery" event on the destination, an unplanned "Proof of Delivery" event is reported on the destination. • **Delivered** ◦ All planned "Delivered" events on the destination are reported. ◦ If there are no planned "Delivered" events on the destination, an unplanned "Delivered" event is reported on the destination. • **Unloading End** ◦ All planned "Unloading End" events on the destination are reported. ◦ If there are no planned "Unloading End" events on the destination, an unplanned "Unloading End" event is reported on the destination. • **Goods Receipt**

Execution Status	Description
	◦ An unplanned "Goods Receipt" is reported. • **Arrival** ◦ If there are no planned "Proof of Delivery", "Delivered" or “Unloading End”events on the destination, all planned "Arrival" events on the destination are reported. ◦ When this freight unit is assigned with tracking units: if there are no planned "Proof of Delivery" events nor "Delivered" events on the destination, all planned "Arrival" events are reported. The following case can also complete a freight unit: • Its **last milestone event** is reported on the destination. Whether an actual event is the last milestone event or not is indicated by the field “**isFinal**”. If it has the field "**isFinal**" whose value is “true”, then it is the last milestone.

Tracking Unit

Execution Status	Description
Not Started	The initial execution status of a tracking unit.
In Execution	If a planned event of the tracking unit is reported, the execution status changes to In Execution.
Completed	When one of the conditions specific to the following event types is met, the tracking unit is considered completed: • **Proof of Delivery** ◦ All planned "Proof of Delivery" events on the destination are reported, or all planned "Proof of Delivery" events are reported. ◦ If there is no planned "Proof of Delivery" event on the destination, an unplanned "Proof of Delivery" event is reported on the destination. • **Delivered** ◦ All planned "Delivered" events on the destination are reported. ◦ If there are no planned "Delivered" events on the destination, an unplanned "Delivered" event is reported on the destination. • **Unloading End** ◦ All planned "Unloading End" events on the destination are reported. ◦ If there are no planned "Unloading End" events on the destination, an unplanned "Unloading End" event is reported on the destination. • **Goods Receipt** ◦ An unplanned "Goods Receipt" is reported. • **Arrival** ◦ If there are no planned "Proof of Delivery", "Delivered" or “Unloading End”events on the destination, all planned "Arrival" events on the destination are reported. The following case can also complete a tracking unit: • Its **last milestone event** is reported. Whether an actual event is the last milestone event or not is indicated by the field “**isFinal**”. If it has the field "**isFinal**" whose value is “true”, then it is the last milestone.

Inbound Delivery

Execution Status	Description
Not Started	The initial execution status of an inbound delivery.
In Execution	If a planned event (except the Goods Receipt event) of the inbound delivery is reported or any of its freight units is in execution, the execution status changes to In Execution.
Completed	When one of the conditions specific to the following event types is met, the inbound delivery is considered completed: • **Goods Receipt** ◦ The planned "Goods Receipt" event of an inbound delivery or all planned "Goods Receipt" event of inbound delivery items are reported. • **Delivered** ◦ The destination's planned "Delivered" event is reported. If it is not planned, a destination's unplanned "Delivered" event is reported. When all freight units of its inbound delivery items are completed, the inbound delivery is also completed.

Outbound Delivery

Execution Status	Description
Not Started	The initial execution status of an outbound delivery.
In Execution	If a planned event of an outbound delivery is reported, the execution status changes to In Execution.
Completed	There are two types of outbound delivery: outbound delivery of a sales order, and of a stock transport order. • **Outbound delivery of a sales order** When one of the conditions specific to the following event types is met, the outbound delivery of a sales order is considered completed: ◦ **Proof of Delivery** ▪ All planned "Outbound Delivery Item POD" events for its outbound delivery items are reported ▪ All planned "Proof of Delivery" events on the destination of the outbound delivery' shipments are reported. ▪ If there is no planned "Proof of Delivery" event on the destination of the outbound delivery's shipments, an unplanned "Proof of Delivery" event is reported on its destination. ◦ **Delivered** ▪ The planned "Delivered" event on the destination is reported. ▪ If there is no planned "Delivered" event on the destination, an unplanned "Delivered" event is reported on the destination. ◦ **Arrival** ▪ If there are no planned "Proof of Delivery", "Delivered" or "Unloading End"events on the destination, all planned "Arrival" events on the destination are reported. The following case can also complete an outbound delivery:

Execution Status	Description
	◦ Its **last milestone event** is reported. Whether an actual event is the last milestone event or not is indicated by the field “isFinal”. If it has the field "isFinal" whose value is “true”, then it is the last milestone. ◦ All freight units of its outbound delivery items are completed. • **Outbound delivery of a stock transport order** When one of the following conditions is met, the outbound delivery of a stock transport order is considered completed: ◦ All planned "Goods Receipt events" for its outbound delivery items are reported ◦ The planned "Goods Receipt" event for the outbound delivery is reported.

Sales Order

Execution Status	Description
Not Started	The initial execution status of a sales order.
In Execution	If one of the sales order items is in execution, the execution status changes to **In Execution**.
Completed	If all the sales order items are completed, the execution status changes to **Completed**.

Sales Order Item

Execution Status	Description
Not Started	The initial execution status of a sales order item.
In Execution	If one of the outbound delivery items is in execution, the execution status changes to **In Execution**.
Completed	If all the outbound delivery items are completed, the execution status changes to **Completed**.

Purchase Order

Execution Status	Description
Not Started	The initial execution status of a purchase order.
In Execution	If one of the purchase order items is in execution, the execution status changes to **In Execution**.
Completed	If all the purchase order items are completed, the execution status changes to **Completed**.

Purchase Order Item

Execution Status	Description
Not Started	The initial execution status of a purchase order item.
In Execution	If one of the inbound or outbound delivery items is in execution or a planned event is reported for the purchase order item, the execution status changes to **In Execution**.
Completed	If all the inbound delivery items or outbound delivery items are completed, the execution status changes to **Completed**. If the purchase order item does not have any deliveries, when the planned "Goods Receipt" event of the purchase order